1.1.1 Let $\Gamma = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle \leq SL_2(\mathbb{Z})$.

Note $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n \in \Gamma$. $\forall n \in \mathbb{Z}$.

i) Let $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, use $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b' \\ c & nc+d \end{pmatrix}$ to show that unless $c=0$, $\exists \gamma \in \Gamma$ $\alpha\gamma$ has bottom row (c, d') with $|d'| \leq |c|/2$.

pf: We need to show $|nc+d| \leq |c|/2$

i.e. $-|c| \leq 2(nc+d) \leq |c|$

$\forall d$ by picking suitable n, $2nc+2d$ lands in $[-|c|, |c|]$.

Precisely, $\frac{2d-|c|}{2|c|} < n < \frac{2d+|c|}{2|c|}$. $n = \left\lfloor \frac{2d+|c|}{2|c|} \right\rfloor$.

ii) Use the identity $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} b & -a \\ d & -c \end{pmatrix}$ to show that this process can be iterated until $\exists \gamma \in \Gamma$ $\alpha\gamma$ has bottom row $(0, *)$.

pf: Let $U_i := \begin{pmatrix} 1 & n_i \\ 0 & 1 \end{pmatrix}$. ~~Then~~ $\alpha_{i+1} = \alpha_i R U_i$ $\alpha_0 := \alpha$.

$R := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ where n_i corresponds to α_i, is equals to $\left\lfloor \frac{2d_i + |c_i|}{2|c_i|} \right\rfloor$

Thus $|c_1| \leq |d_0|/2$ $|d_2| \leq |c_1|/2 \leq |d_0|/4$.

Iteration gives $|d_n| \leq |d_0|/2^n$. As $d_i \in \mathbb{Z}$,

$\exists N$ $d_N = 0$. By doing one more R, $\alpha_N R = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$.

iii) Show that infact bottom row is $(0, \pm 1)$.

Since $\alpha_N R \in SL_2(\mathbb{Z})$. Denote $\alpha_N R = \begin{pmatrix} x_1 & x_2 \\ 0 & x_4 \end{pmatrix}$.

then $x_1 x_4 = 1$. Thus $x_1, x_4 = \pm 1$.

(1.1.1) iv) Since $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = -I$ it can be taken to be $(0,1)$,

v) Show that therefore $\alpha\gamma \in \Gamma$. and so $\alpha \in \Gamma$.

Pf: Since $\alpha_N R = \begin{pmatrix} x_1 & x_2 \\ 0 & 1 \end{pmatrix}$, $x_1 = 1$. $x_2 = n$ for some $n \in \mathbb{Z}$.

Thus $\alpha(RU_1 RU_2 \cdots RU_N R) = \alpha_N R = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \in \Gamma$.

$U_i, R \in \Gamma \Rightarrow \alpha \in \Gamma$.

vi) Thus Γ is all of $SL_2(\mathbb{Z})$.

Pf: α is picked arbitrarily.

1.1.2 (a) Show $\operatorname{Im} \gamma(\tau) = \operatorname{Im} \tau / |c\tau + d|^2$ $\forall \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2\mathbb{Z}$

Pf: $\operatorname{Im} \gamma(\tau) = \operatorname{Im} \dfrac{a\tau + b}{c\tau + d} = \operatorname{Im} \dfrac{(a\tau+b)(c\bar{\tau}+d)}{|c\tau+d|^2}$

$= \operatorname{Im} (ac|\tau|^2 + ad\tau + bc\bar{\tau} + bd) / |c\tau + d|^2$

$= (ad - bc) \operatorname{Im} \tau / |c\tau + d|^2 = \operatorname{Im} \tau / |c\tau + d|^2$.

(b) Show $(\gamma\gamma')(\tau) = \gamma(\gamma'(\tau))$ $\forall \gamma, \gamma' \in SL_2(\mathbb{Z})$, $\tau \in \mathcal{H}$.

Pf: $(\gamma\gamma')(\tau) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}(\tau) = \dfrac{(aa' + bc')\tau + ab' + bd'}{(ca' + dc')\tau + cb' + cd'}$.

$\gamma(\gamma'(\tau)) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \dfrac{a'\tau + b'}{c'\tau + d'} = \dfrac{a\dfrac{a'\tau + b'}{c'\tau + d'} + b}{c\dfrac{a'\tau + b'}{c'\tau + d'} + d}$

$= \dfrac{aa'\tau + ab' + bc'\tau + bd'}{ca'\tau + cb' + dc'\tau + dd'}$ Thus equal.

(c) Show $d\gamma(\tau)/d\tau = 1/(c\tau + d)^2$ $\forall \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$.

Pf: $d\gamma(\tau)/d\tau = \dfrac{a(c\tau + d) - (a\tau + b)c}{(c\tau + d)^2} = 1/(c\tau + d)^2$.

1.1.3. (a) Show $M_k(SL_2(\mathbb{Z}))$ forms a $\mathbb{C}$-vector space

Pf: Abelian group: f, g holomorphic ~~at~~ in $\mathcal{H}$, at ∞

$\Rightarrow f-g$ holomorphic in $\mathcal{H}$, at ∞.

f, g weight $k \Rightarrow (f-g)(\gamma(\tau)) = (c\tau+d)^k (f-g)(\tau)$.

scalar product: f holomorphic in $\mathcal{H}$, at ∞

$\Rightarrow zf$ holomorphic in $\mathcal{H}$, at ∞

f weight $k \Rightarrow zf(\gamma(\tau)) = z(c\tau+d)^k (zf)(\tau)$

(b) $f \in M_k(SL_2(\mathbb{Z}))$, $g \in M_l(SL_2(\mathbb{Z}))$. Show $fg \in M_{k+l}(SL_2(\mathbb{Z}))$

Pf: $f \cdot g$ holomorphic in $\mathcal{H}$. $\lim_{z \to 0} \frac{fg(\tau+z) - fg(\tau)}{z} \in \mathbb{C}$ $\forall \tau \in \mathcal{H}$

f, g holomorphic at $\infty \Rightarrow$ $f = a_0 + a_1 q + \cdots$
$g = b_0 + b_1 q + \cdots$

$\Rightarrow f \cdot g = c_0 + c_1 q + \cdots$ holomorphic at ∞.

$$(f \cdot g)(\gamma(\tau)) = f \cancel{(c\tau+d)^{-k}(\gamma(\tau))} \cdot g(\gamma(\tau))$$
$$= (c\tau+d)^{-k} f(\tau) \cdot (c\tau+d)^{-l} g(\tau).$$
$$= (c\tau+d)^{-(k+l)} fg(\tau).$$

(c) Show $S_k(SL_2(\mathbb{Z})) \subseteq M_k(SL_2(\mathbb{Z}))$ as ~~sub~~ $\mathbb{C}$-vector subspace

and $S(SL_2(\mathbb{Z})) \trianglelefteq M(SL_2(\mathbb{Z}))$.

Pf: $\forall f \in S_k(SL_2(\mathbb{Z}))$ $f(\tau) = \sum_{n=1}^{\infty} a_n q^n \overset{a_0=0}{=} \sum_{n=0}^{\infty} a_n q^n \in M_k(SL_2(\mathbb{Z}))$

$\forall G \in M(SL_2(\mathbb{Z}))$, $F \in S(SL_2(\mathbb{Z}))$,

$$G \cdot F(\tau) = \left(\sum_{n=0}^{\infty} a_n q^n\right)\left(\sum_{n=1}^{\infty} b_n q^n\right) = \sum_{m=1}^{\infty} \sum_{i+j=m} a_i b_j q^n$$

$G(\tau) \cdot F(\tau) =$

$\in S(SL_2(\mathbb{Z}))$.

1.1.4. Let $k \geq 3$, $L' = \mathbb{Z}^2 - \{(0,0)\}$.

(a) Show $\displaystyle\sum_{(c,d)\in L'} (\sup\{|c|,|d|\})^{-k}$ converges.

by considering partial sums over expanding squares.

Pf: Denote $S_k(X) := \displaystyle\sum_{(c,d)\in X} (\sup\{|c|,|d|\})^{-k}$. $X \subset L'$.

$L_n := \{(c,d)\in L' : \sup\{|c|,|d|\} = n\}$. $n \geq 1$.

$$S_k(L_n) = \sum_{(c,d)\in L_n} n^{-k} = \frac{|L_n|}{n^k} = \frac{8n}{n^k} = \frac{8}{n^{k-1}}.$$

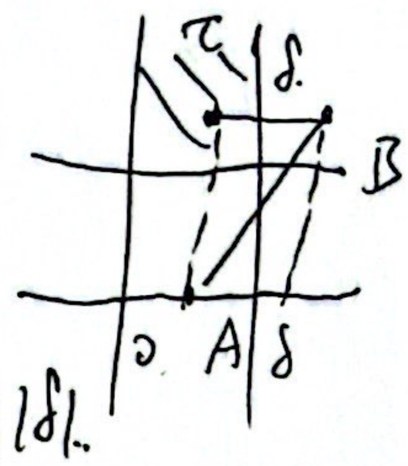
Thus $S_k(L') = \displaystyle\sum_{n=1}^{\infty} S_k(L_n) = \sum_{n=1}^{\infty} \frac{8}{n^{k-1}} < \infty$ since $k \geq 3$.

$= 8\zeta(k-1)$.

(b) Fix A, B. Let $\Omega = \{\tau \in \mathcal{H} : |\mathrm{Re}\,\tau| \leq A,\ \mathrm{Im}\,\tau \geq B\}$.

Prove there is $C > 0$ s.t. $|\tau + \delta| > C \sup\{1, |\delta|\}$ $\forall \tau \in \Omega$, $\delta \in \mathbb{R}$.

Pf: ① If $|\delta| < B$, say $|\delta| \leq kB$



triangle inequality tells

$$|\tau + \delta| \geq |\tau| - |\delta| \geq B - |\delta| \geq \left(\frac{1-k}{k}\right)|\delta|.$$

Thus we could let $C = \dfrac{1-k}{k}$.

② If for the same $k = \dfrac{1}{C+1}$, $|\delta| \geq kB$

1.1.5. Show $\frac{1}{\tau} + \sum_{d=1}^{\infty}\left(\frac{1}{\tau-d} + \frac{1}{\tau+d}\right) = \pi\cot\pi\tau = \pi i - 2\pi i \sum_{m=0}^{\infty} q^m$.

Pf: LHS: Recall $\sin\pi\tau = \pi\tau \prod_{n=1}^{\infty}\left(1-\frac{\tau^2}{n^2}\right)$.

Then $\log\sin\pi\tau = \log\pi + \log\tau + \sum_{n=1}^{\infty}\log\left(1-\frac{\tau^2}{n^2}\right)$

Take derivative wrt τ, gives

$$\pi\cot\pi\tau = \frac{1}{\tau} + \sum_{n=1}^{\infty}\frac{-\frac{2\tau}{n^2}}{1-\frac{\tau^2}{n^2}} = \frac{1}{\tau} + \sum_{n=1}^{\infty}\left(\frac{1}{\tau-n}+\frac{1}{\tau+n}\right).$$

RHS: $\sin\pi\tau = \frac{e^{i\pi\tau}-e^{-i\pi\tau}}{2i}$, $\cos\pi\tau = \frac{e^{i\pi\tau}+e^{-i\pi\tau}}{2}$

Thus $\pi\cot\pi\tau = \pi i\,\frac{e^{i\pi\tau}+e^{-i\pi\tau}}{e^{i\pi\tau}-e^{-i\pi\tau}} = \pi i\,\frac{e^{2\pi i\tau}+1}{e^{2\pi i\tau}-1}$

$$= \pi i\left(1 - 2\frac{1}{1-e^{2\pi i\tau}}\right) = \pi i - 2\pi i\sum_{m=0}^{\infty} e^{2\pi i\tau m}.$$

1.1.7. Bernoulli numbers B_k are defined by $\frac{t}{e^t-1} = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!}$

Thus they are calculable in succession by matching coefficients in the power series identity.

$$t = (e^t-1)\sum_{k=0}^{\infty} B_k \frac{t^k}{k!} = \sum_{n=1}^{\infty}\left(\sum_{k=0}^{n-1}\binom{n}{k}B_k\right)\frac{t^n}{n!}$$

and they are rational. Since $\frac{t}{e^t-1} + \frac{t}{2} = \frac{t}{2}\cdot\frac{e^t+1}{e^t-1}$ is even, it follows that $B_1 = -\frac{1}{2}$, $B_k = 0$ $\forall k$ odd.

(a) Show $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$.

Sol:

$$\binom{1}{0}B_0 = 1. \Rightarrow B_0 = 1.$$

$$\frac{1}{2!}\left(\binom{2}{0}B_0 + \binom{2}{1}B_1\right) = 0 \Rightarrow B_1 = -\frac{1}{2}.$$

$$\frac{1}{3!}\left(\binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2\right) = 0 \Rightarrow B_2 = \frac{1}{6}$$

$$\binom{4}{0}B_0 + \binom{4}{1}B_1 + \binom{4}{2}B_2 + \binom{4}{3}B_3 = 0 \Rightarrow B_3 = 0.$$

$$\binom{5}{0}B_0 + \binom{5}{1}B_1 + \binom{5}{2}B_2 + \binom{5}{3}B_3 + \binom{5}{4}B_4 = 0 \Rightarrow B_4 = -\frac{1}{30}$$

$$\binom{7}{0}B_0 + \binom{7}{1}B_1 + \binom{7}{2}B_2 + \cancel{\binom{7}{3}B_3} + \binom{7}{4}B_4 + \cancel{\binom{7}{5}}\binom{7}{6}B_6 = 0 \Rightarrow B_6 = \frac{1}{42}$$

(b) i) Use $\frac{1}{\tau} + \sum_{d=1}^{\infty}\left(\frac{1}{\tau-d} + \frac{1}{\tau+d}\right) = \pi\cot\pi\tau = \pi i - 2\pi i\sum_{m=0}^{\infty} q^m$, $q = e^{2\pi i\tau}$

to show $$1 - 2\sum_{k=1}^{\infty}\zeta(2k)\tau^{2k} = \pi\tau\cot\pi\tau = \pi i\tau + \sum_{k=0}^{\infty} B_k\frac{(2\pi i\tau)^k}{k!}$$

Pf: RHS: to prove $$-2\pi i\tau\sum_{m=0}^{\infty} q^m = \sum_{k=0}^{\infty} B_k\frac{(2\pi i\tau)^k}{k!} = \frac{2\pi i\tau}{e^{2\pi i\tau}-1}$$

to prove $$\sum_{m=0}^{\infty} e^{2\pi i\tau m} = \frac{1}{1-e^{2\pi i\tau}}, \quad \frac{1}{1-x} = \sum_{m=0}^{\infty} x^m$$

LHS: $$\tau\sum_{d=1}^{\infty}\left(\frac{1}{\tau-d} + \frac{1}{\tau+d}\right) = 2\sum_{d=1}^{\infty}\frac{\tau^2}{\tau^2-d^2} = -2\sum_{d=1}^{\infty}\frac{\tau^2}{d^2}\frac{1}{1-\left(\frac{\tau}{d}\right)^2} = -2\sum_{d=1}^{\infty}\frac{\tau^2}{d^2}\sum_{k=0}^{\infty}\left(\frac{\tau}{d}\right)^{2k}$$

$$= -2\sum_{d=1}^{\infty}\sum_{k=0}^{\infty}\frac{\tau^{2(k+1)}}{d^{2(k+1)}} = -2\sum_{d=1}^{\infty}\sum_{k=1}^{\infty}\frac{\tau^{2k}}{d^{2k}} = -2\sum_{k=1}^{\infty}\zeta(2k)\tau^{2k}$$

(1.1.7) ii') Use these to show $\forall k \geq 2$ even,

$$2\zeta(k) = -\frac{(2\pi i)^k}{k!} B_k .$$

Pf: From (b) i), $1 - 2\sum_{k=1}^{\infty} \zeta(2k)\tau^{2k} = \pi i \tau + \sum_{k=0}^{\infty} B_k \frac{(2\pi i \tau)^k}{k!}$

Then $-2\sum_{k=1}^{\infty} \zeta(2k)\tau^{2k} = \pi i \tau + \sum_{k=1}^{\infty} B_k \frac{(2\pi i)^k}{k!} \tau^k$.

$$= \sum_{k=1}^{\infty} B_{2k} \frac{(2\pi i)^{2k}}{(2k)!} \tau^{2k} .$$

Thus $-2\zeta(2k) = B_{2k} \frac{(2\pi i)^{2k}}{(2k)!}$

i.e. $2\zeta(k) = -\frac{(2\pi i)^k}{k!} B_k \quad \forall k \geq 2$ even.

iii) In particular, $\zeta(2) = \frac{\pi^2}{6}$, $\zeta(4) = \frac{\pi^4}{90}$, $\zeta(6) = \frac{\pi^6}{945}$.

Pf: $2\zeta(2) = -\frac{(2\pi i)^2}{2} B_2 = \frac{\pi^2}{3} \Rightarrow \zeta(2) = \frac{\pi^2}{6}$.

$2\zeta(4) = -\frac{(2\pi i)^4}{4!} B_4 = \frac{\pi^4}{45} \Rightarrow \zeta(4) = \frac{\pi^4}{90}$.

$2\zeta(6) = -\frac{(2\pi i)^6}{6!} B_6 = \frac{2\pi^6}{945} \Rightarrow \zeta(6) = \frac{\pi^6}{945}$.

iv) Also, this shows that normalized Eisenstein series of weight k has rational coefficients with a common denominator,

$$E_k(\tau) = \frac{G_k(\tau)}{2\zeta(k)} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n .$$

Pf: There is $G_k(\tau) = 2\zeta(k) + 2\frac{(2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad k > 2$ even

Then $E_k(\tau) = 1 + \frac{(2\pi i)^k}{\zeta(k)(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$

$$= 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n .$$

(1.1.7) (c) Equate coefficients in $E_8(\tau) = E_4^2(\tau)$ to establish (1.3) $\sigma_7(n) = \sigma_3(n) + 120 \sum_{i=1}^{n-1} \sigma_3(i)\,\sigma_3(n-i)$

Sol: $E_8(\tau) = 1 - \frac{16}{B_8} \sum_{n=1}^{\infty} \sigma_7(n) q^n$

$$E_4^2(\tau) = \left(1 - \frac{8}{B_4} \sum_{n=1}^{\infty} \sigma_3(n) q^n\right)^2$$

$$= 1 - \frac{16}{B_4} \sum_{n=1}^{\infty} \sigma_3(n) q^n + \frac{64}{B_4^2} \sum_{n=1}^{\infty} \sum_{m=1}^{n-1} \left(\sigma_3(m) \cdot \sigma_3(n-m)\right) q^n$$

Thus $-\frac{16}{B_8} \sigma_7(n) = -\frac{16}{B_4} \sigma_3(n) + \frac{64}{B_4^2} \sum_{m=1}^{n-1} \left(\sigma_3(m) \cdot \sigma_3(n-m)\right)$ $\forall n = 1, 2, \ldots$

$\binom{9}{0} B_0 + \binom{9}{1} B_1 + \binom{9}{2} B_2 + \binom{9}{4} B_4 + \binom{9}{6} B_6 + \binom{9}{8} B_8 = 0 \Rightarrow B_8 = -\frac{1}{30}$.

Thus $-\sigma_7(n) = -\sigma_3(n) + \frac{4}{B_4} \sum_{m=1}^{n-1} \sigma_3(m)\,\sigma_3(n-m)$

$$\Rightarrow \sigma_7(n) = \sigma_3(n) - 120 \sum_{m=1}^{n-1} \sigma_3(m)\,\sigma_3(n-m).$$

(d) Show that $\Delta(\tau) = g_2^3(\tau) - 27 g_3^2(\tau)$ has expansion with coefficient $a_0 = 0$, $a_1 = (2\pi)^{12}$.

Pf: $\Delta(\tau) = (60 G_4(\tau))^3 - 27 (140 G_6(\tau))^2$

$= \cancel{216000}\ \cancel{G_4^3(\tau) - 1029}$.

$$60^3 \left(2\zeta(4) + 2\frac{(2\pi i)^4}{3!} \sum_{n=1}^{\infty} \sigma_3(n) q^n\right)^3 - \cancel{\frac{27 \cdot 140^2}{4}}\left(2\zeta(6) + 2\frac{(2\pi i)^6}{5!} \sum_{n=1}^{\infty} \sigma_5(n) q^n\right)^2$$

$a_0 = \cancel{216000}\ 60^3 \cdot (2\zeta(4))^3 - \cancel{\frac{27 \cdot 140^2}{4}} (2\zeta(6))^2 = \frac{2^6}{3^3}\pi^{12} - 3^3 \cdot 140^2 \cdot 2^2 \frac{\pi^{12}}{945^2}$

$= 0.$

$$a_1 = \left(60^3 \cdot 2\frac{(2\pi i)^4}{3!} \sigma_3(1)\right)^3 - 27 \cdot 140^2 \cdot 2 \cdot \frac{(2\pi i)^6}{5!} \sigma_5(1) \cdot 2 \cdot (2\zeta(6))^2$$

$$\cdot 3 \cdot (2\zeta(4))^2$$

$$= \left(60^3 2^2 \frac{1}{90^2} 2^4 + \frac{9}{5} 140^2 \frac{2^6}{945}\right)\pi^{12} = \left(\frac{5}{3} \cdot 2^{10} + \frac{7}{3} 2^{10}\right)\pi^{12}$$

$= (2\pi)^{12}$.